

CHAPTER 6

6.1) DERIVATIVE OF FERMI-DIRAC FUNCTION

$$f(\varepsilon) = \{\exp[(\varepsilon - \mu)/\tau] + 1\}^{-1} ;$$

$$(\partial f / \partial \varepsilon)_{\mu, \tau} = -\{\exp[(\varepsilon - \mu)/\tau] + 1\}^{-2} \exp[(\varepsilon - \mu)/\tau] (1/\tau) ;$$

when $\varepsilon = \mu$, then $\exp[(\varepsilon - \mu)/\tau] = 1$, $-(\partial f / \partial \varepsilon)_{\mu, \tau} = 1/4\tau$.

6.2) SYMMETRY OF FILLED AND VACANT ORBITALS

$$f(\mu - \delta) = \frac{1}{\exp[(\mu - \delta - \mu)/\tau] + 1} = \frac{1}{\exp(-\delta/\tau) + 1} = \frac{\exp(\delta/\tau)}{\exp(\delta/\tau) + 1} ;$$

$$f(\mu + \delta) = \frac{1}{\exp[(\mu + \delta - \mu)/\tau] + 1} = \frac{1}{\exp(\delta/\tau) + 1} = 1 - f(\mu - \delta) .$$

Here $1 - f(\mu - \delta)$ is the probability that an orbital at $\varepsilon = \mu - \delta$, an energy δ below the Fermi level, is vacant. In semiconductors such an orbital is said to be occupied by a hole.

6.3) DISTRIBUTION FUNCTION FOR DOUBLE OCCUPANCY STATISTICS

(a) Occupancy: 0 1 2

Energy : 0 ε 2ε

$$\mathcal{Z}_{\text{DO}} = 1 + \lambda e^{-\varepsilon/\tau} + \lambda^2 e^{-2\varepsilon/\tau}$$

$$\langle N \rangle_{\text{DO}} = \frac{1}{\mathcal{Z}} [0 + \lambda e^{-\varepsilon/\tau} + 2\lambda^2 e^{-2\varepsilon/\tau}] .$$

(b) Occupancy: 0,0 0,1 1,0 1,1

Energy : 0 ε ε 2ε

$$\mathcal{Z} = 1 + 2\lambda e^{-\varepsilon/\tau} + \lambda^2 e^{-2\varepsilon/\tau}$$

$$= [1 + \lambda e^{-\varepsilon/\tau}]^2 ;$$

$$\langle N \rangle = \frac{1}{\mathcal{Z}} [0 + 2\lambda e^{-\varepsilon/\tau} + 2\lambda^2 e^{-2\varepsilon/\tau}] .$$

The last line may be written

$$\langle N \rangle = 2\lambda e^{-\varepsilon/\tau} \frac{1 + \lambda e^{-\varepsilon/\tau}}{[1 + \lambda e^{-\varepsilon/\tau}]^2} = \frac{2}{\exp[(\varepsilon - \mu)/\tau] + 1} \quad .$$

The Gibbs sum is square of a single-orbital Gibbs sum, as one would expect for two independent single-orbital systems. See Problem 3.9 for comparison. The occupancy is just twice the Fermi-Dirac distribution function.

6.4) ENERGY OF GAS OF EXTREME RELATIVISTIC PARTICLES

$\varepsilon = pc = (2\pi\hbar/\lambda)c = (\pi\hbar c/L)n$. Here we first used $p = 2\pi\hbar/\lambda$ and then $L = n\lambda/2$, where L is the size of a cube of volume $V = L^3$ assumed to contain the particle. The partition function for one particle is converted to an integral as for photons; see (4.17):

$$\begin{aligned} Z = Z_1 &= \sum \exp[-(\pi\hbar c/\tau L)n] = (\pi/2) \int_0^\infty \exp[-(\pi\hbar c/\tau L)n] n^2 dn \\ &= (\pi/2)(\tau L/\pi\hbar c)^3 \int_0^\infty e^{-x} x^2 dx = \text{constant} \times \tau^3 \quad , \end{aligned}$$

where $x = (\pi\hbar c/\tau L)n$. From $Z \propto \tau^3$:

$$U = \tau^2 \frac{\partial}{\partial \tau} \log Z = 3\tau, \text{ per particle.}$$

6.5) INTEGRATION OF THERMODYNAMIC IDENTITY FOR AN IDEAL GAS

From $U = 3N\tau/2$:

$$(\partial U/\partial \tau)_V = 3N/2 = C_V, \quad (\partial U/\partial V)_\tau = 0 \quad .$$

If these and $p = n\tau/V$ are inserted into (75):

$$\begin{aligned} d\sigma &= C_V d\tau/\tau + NdV/V = C_V d(\log \tau) + N d(\log V) \quad ; \\ \sigma &= C_V \log \tau + N \log V + \text{constant.} \end{aligned} \quad (76)$$

6.6) ENTROPY OF MIXING

For a single gas of N atoms with the concentration $n = N/V$, from (34):

$$\sigma = N[\log(n_Q/n) + 5/2] \quad .$$

For two distinguishable independent gases, each state of gas A may be combined with each state of gas B to generate a distinct state of the combined system. If g_A and g_B are the numbers of states accessible to each gas, $g = g_A g_B$ is the number of states accessible to the two-gas system, and the two-gas entropy is the sum of the one-gas entropies: $\sigma = \sigma_A + \sigma_B$.

This is true both before and after diffusive contact. Before contact, each of the two concentrations is $n = n_i = N/V$, hence

$$\sigma_i = N[\log(n_{QA}/n_i) + \log(n_{QB}/n_i) + 5] \quad ,$$

where n_{QA} and n_{QB} will in general be different. After diffusive equilibrium, $n = n_f = N/2V = n_i/2$; therefore

$$\begin{aligned} \sigma_f &= N[\log(n_{QA}/n_f) + \log(n_{QB}/n_f) + 5] \\ &= N[\log(2n_{QA}/n_i) + \log(2n_{QB}/n_i) + 5] \\ &= \sigma_i + 2N \log 2 \quad . \end{aligned}$$

The additional term $2N \log 2$ may be understood as follows. For every A-atom orbital in volume V_A there is an equivalent A-atom orbital in V_B , and for every B-atom orbital in V_B there is an equivalent B-atom orbital in V_A . States in which these equivalent orbitals are occupied are initially inaccessible, but they become accessible upon diffusive contact. Every accessible state of the two-gas system with diffusive contact may be viewed as having been generated from one of the initially accessible states by interchanging the occupancies of an arbitrary number (from 0 to $2N$) of equivalent orbital pairs. For $2N$ occupied

orbitals, there are 2^{2N} distinguishable combinations of interchanges, leading to 2^{2N} distinguishable accessible states for every initially accessible state. The entropy of mixing is the logarithm of this multiplicity of mixing: $\sigma_M = \log 2^{2N} = 2N \log 2$.

If the particles are indistinguishable, these occupancy interchanges do not lead to new distinguishable states but merely re-arrange the accessible states amongst each other. The two gases in diffusive contact are then simply a single gas of $2N$ atoms with the unchanged concentration $n_f = 2N/2V = n_i$, and with the entropy

$$\sigma_f = 2N[\log(n_Q/n_i) + 5/2] \quad .$$

For indistinguishable gases $n_{QA} = n_{QB}$; therefore $\sigma_f = \sigma_i$.

Comment. Students often ask whether particles can be "nearly" identical. The answer is: no! Particles differ either by a finite discrete amount or not at all. For example, the different isotopes of an element differ in the (discrete) number of neutrons in their nuclei.

6.7) RELATION OF PRESSURE AND ENERGY DENSITY

(a) From the discussion in Chapter 3 -- see (3.25) -- and in Appendix D, the pressure p_s in state s is $p_s = -(\partial \epsilon_s / \partial V)_N$. We have written $(\partial \epsilon_s / \partial V)_N$ rather than $d\epsilon_s/dV$ to indicate that the number of particles in the system is kept constant during the volume change. The thermal average pressure is then

$$p \equiv \langle p \rangle = \sum_s p_s P(\epsilon_s) = - \frac{1}{Z} \sum_s (\partial \epsilon_s / \partial V)_N \exp(-\epsilon_s / \tau) \quad ,$$

where $P(\epsilon_s)$ is the probability (3.11) of finding the system in state s .

(b) The energy ϵ_s of each state of the system is the sum over the energies of all occupied orbitals. According to

(3.59), the energies of all orbitals depend on the volume through the common factor $L^{-2} = V^{-2/3}$, hence

$$\varepsilon_s \propto V^{-2/3} ; \quad \log \varepsilon_s = -\frac{2}{3} \log V + \text{constant};$$

$$\frac{1}{\varepsilon_s} \left(\frac{\partial \varepsilon_s}{\partial V} \right)_N = -\frac{2}{3} \frac{1}{V} \text{ or } \left(\frac{\partial \varepsilon_s}{\partial V} \right)_N = -\frac{2}{3} \frac{\varepsilon_s}{V} .$$

(c) From (a) and (b), with (3.12),

$$p = \frac{2}{3V} \frac{1}{Z} \sum_s \varepsilon_s \exp(-\varepsilon_s/\tau) = 2U/3V .$$

6.8) TIME FOR A LARGE FLUCTUATION

(a) At $T \cong 300$ K and $p = 1$ atm $\cong 1.013 \times 10^6$ dyne cm^{-2} we have $n = p/k_B T = 2.45 \times 10^{19} \text{cm}^{-3}$. For ${}^4\text{He}$ at $T = 300$ K, we have $n_Q \cong 7.81 \times 10^{24} \text{cm}^{-3}$. Hence for 10^2cm^3 of ${}^4\text{He}$ gas:

$$\begin{aligned} \sigma &= nV[\log(n_Q/n) + 5/2] \\ &= 2.45 \times 10^{21} \times [\log(7.81 \times 10^{24}/2.45 \times 10^{19}) + 5/2] \\ &= 3.71 \times 10^{22} , \end{aligned}$$

$$g_1 = \exp(3.71 \times 10^{22}) .$$

(b) At $V = 0.05$ liter and at the same temperature, the concentration is doubled, n_Q remains unchanged. The entropy is therefore reduced by $\Delta\sigma = -nV \log 2 \cong -1.70 \times 10^{21}$, to $\sigma \cong 3.54 \times 10^{22}$ which implies $g_2 \cong \exp(3.54 \times 10^{21})$.

$$(c) \quad g_2/g_1 = \exp(+\Delta\sigma) = \exp(-1.70 \times 10^{21}) \cong 10^{-7.36 \times 10^{20}} .$$

(d) There are about $\pi \times 10^7$ sec in a year (see solution to Problem 4.4), hence the collision frequency is

$$f_c \cong 2.7 \times 10^{21} \times 10^{10} \times \pi \times 10^7 \text{yr}^{-1} \cong 8.5 \times 10^{38} \text{yr}^{-1} .$$

(e) The result (c) states, in effect, that only one out of every

$$g_1/g_2 \cong 10^{7 \times 10^{20}}$$

states is a state for which all atoms are in one-half of the volume. Hence only once every (g_1/g_2) collisions can we expect such a state to occur. If there are $f_c \cong 10^{39}$ collisions per year, this will take place once in each time interval

$$t = (g_1/g_2)/f_c \cong 10^{7 \times 10^{20}} / 10^{39} \text{ yr}^{-1} \cong 10^{7 \times 10^{21}} \text{ years.}$$

This means it will never occur.

6.9) GAS OF ATOMS WITH INTERNAL DEGREE OF FREEDOM

From (44), $Z_{\text{int}} = 1 + \exp(-\Delta/\tau)$, independent of the volume. The various results below are expressed most conveniently by defining

$$n_Q^* = n_Q Z_{\text{int}} = n_Q [1 + \exp(-\Delta/\tau)].$$

(a) From (48):

$$\mu = \tau \log(n/n_Q^*).$$

(b) From (24) and (49):

$$\begin{aligned} F &= F_{\text{ext}} + F_{\text{int}} = N\tau [\log(n/n_Q) - 1] - N\tau \log Z_{\text{int}} \\ &= N\tau [\log(n/n_Q^*) - 1]. \end{aligned}$$

Note that the results for both μ and F are obtained from those for the gas without internal degrees of freedom, by simply substituting $n_Q \rightarrow n_Q^*$.

(c) From (50), (49) and (34):

$$\sigma_{\text{int}} = \frac{\partial}{\partial \tau} (N\tau \log Z_{\text{int}}) = N \log Z_{\text{int}} + \frac{N\Delta}{\tau} \frac{1}{\exp(\Delta/\tau) - 1},$$

$$\sigma = \sigma_{\text{ext}} + \sigma_{\text{int}} = N \log(n_Q^*/n) + 5/2 + \frac{\Delta/\tau}{\exp(\Delta/\tau) - 1}$$

Note that this is not the same as simply replacing n_Q by n_Q^* in the Sackur-Tetrode equation (34).

(d) The contribution F_{int} to F does not depend on the volume. Hence, from (28) and (29),

$$p = - (\partial F / \partial V)_{\tau, N} = - (\partial F_{\text{ext}} / \partial V)_{\tau, N} = N\tau / V ,$$

as for the gas without internal degrees of freedom.

(e) From $F = U - \tau\sigma$ we have the contribution to the energy of the system, due to the internal degrees of freedom:

$$U_{\text{int}} = F_{\text{int}} + \tau\sigma_{\text{int}} = \frac{N\Delta}{\exp(\Delta/\tau)+1} .$$

This depends on neither pressure nor volume. Hence, the heat capacity is increased by

$$C_{\text{int}} = \frac{d}{d\tau} U_{\text{int}} = \frac{N\Delta^2}{2} \frac{\exp(\Delta/\tau)}{[\exp(\Delta/\tau)+1]^2} .$$

With (38c):

$$C_p = \frac{5}{2}N + C_{\text{int}} .$$

The additional term vanishes both at low and at high temperatures. At low temperatures ($\tau \ll \Delta$) the internal degree of freedom is not excited; at high temperature ($\tau \gg \Delta$), both internal states have equal probability to be excited, with an average energy of internal excitation of $\Delta/2$, independent of τ . At intermediate temperatures there is a bump (Schottky anomaly) in the heat capacity. The maximum heat capacity occurs when

$$\frac{\Delta}{\tau} = \frac{\exp(\Delta/\tau)+1}{\exp(\Delta/\tau)-1} ,$$

an equation easily solved by iteration on a programmable calculator, with the solution $\Delta/\tau \cong 2.40$ or $\tau \cong 0.417\Delta$. The additional heat capacity at that temperature is 0.439, in fundamental units.

6.10) ISENTROPIC RELATIONS OF IDEAL GAS

Because $n_Q \propto \tau^{3/2}$ we may write the Sackur-Tetrode equation (34) in the form (61),

$$\sigma(\tau, V) = N (\log \tau^{3/2} + \log V + \text{constant}) + \sigma_{\text{int}}(\tau) \quad , \quad (S1)$$

where the constant depends on neither τ nor V and where we have added the entropy of the internal degrees of freedom of the gas atoms, which depends on τ but not on V . We obtain $\sigma(\tau, p)$ by setting $V = N\tau/p$ in (S1), from the ideal gas law, which remains valid even in the presence of internal degrees of freedom (see solution to preceding problem):

$$\sigma(\tau, p) = N(\log \tau^{5/2} - \log p + \text{constant}) + \sigma_{\text{int}}(\tau) \quad . \quad (S2)$$

From (35) and (37), with (S1) and (S2):

$$C_V = \frac{3}{2}N + \tau(d\sigma_{\text{int}}/d\tau) = \frac{3}{2}N + C_{\text{int}} \quad , \quad (S3)$$

$$C_p = \frac{5}{2}N + C_{\text{int}} \quad , \quad (S4)$$

$$C_p - C_V = (\gamma-1)C_V = \frac{\gamma-1}{\gamma} C_p = N \quad , \quad (S5)$$

where $\gamma \equiv C_p/C_V$.

(a) In an isentropic expansion, from (S1):

$$d\sigma = \frac{3N}{2} \frac{d\tau}{\tau} + N \frac{dV}{V} + C_{\text{int}} \frac{d\tau}{\tau} = C_V \frac{d\tau}{\tau} + N \frac{dV}{V} = 0 \quad ,$$

or, with (S5),

$$\frac{d\tau}{\tau} = - \frac{N}{C_V} \frac{dV}{V} = - (\gamma-1) \frac{dV}{V} \quad , \quad (S6)$$

as claimed in the second equation (80). We may write (S6) as

$$d \log(\tau V^{\gamma-1}) = 0 \quad ,$$

which implies (66). Alternately, from (S2), with the aid of (S5):

$$\begin{aligned} d\sigma &= \frac{5N}{2} \frac{d\tau}{\tau} - N \frac{dp}{p} + C_{int} \frac{d\tau}{\tau} = C_p \frac{d\tau}{\tau} - N \frac{dp}{p} = 0 \quad , \\ \frac{dp}{p} &= \frac{C_p}{N} \frac{d\tau}{\tau} = \frac{\gamma}{\gamma-1} \frac{d\tau}{\tau} = -\gamma \frac{dV}{V} \quad , \end{aligned} \quad (S7)$$

which contains the remaining two equations in (80). They may also be written

$$d \log(p\tau^{\gamma/(1-\gamma)}) = 0 \quad \text{and} \quad d \log(pV^{\gamma}) = 0 \quad ,$$

which imply (67) and (68).

(b) Inasmuch as (S7) refers to an isentropic process we may write it in the form

$$\frac{dp}{dV} = \left(\frac{\partial p}{\partial V} \right)_{\sigma} = -\gamma \frac{p}{V} \quad , \quad B_{\sigma} = + \gamma p \quad .$$

For an isothermal change, $p = N\tau/V$:

$$(\partial p / \partial V)_{\tau} = -N\tau/V^2 = -p/V \quad , \quad B_{\tau} = p \quad .$$

The application to the velocity of sound in a gas is discussed in Chapter 15.

6.11) CONVECTIVE ISENTROPIC EQUILIBRIUM OF THE ATMOSPHERE

(a) Mechanical equilibrium means that the pressure difference between $z+dz$ and z equals the weight per unit area of the gas contained in that interval,

$$\frac{dp}{dz} = -\rho g = -Mgn = -Mgp/\tau \quad .$$

Here M is the mass of the molecules, g the gravitational acceleration, and we have used the ideal gas law to replace $n = N/V$ by p/τ . Next,

$$\begin{aligned} \frac{dT}{dz} &= \frac{1}{k_B} \frac{d\tau}{dz} = \frac{1}{k_B} \left(\frac{\partial \tau}{\partial p} \right)_\sigma \frac{dp}{dz} = -\frac{Mg}{k_B} \frac{p}{\tau} \left(\frac{\partial \tau}{\partial p} \right)_\sigma = -\frac{Mg}{k_B} \left(\frac{\partial \log \tau}{\partial \log p} \right)_\sigma \\ &= -\frac{Mg}{k_B} \frac{\gamma-1}{\gamma}, \end{aligned} \quad (S1)$$

where in the last equality we have used (67) or the third equation (80). (See also solution to preceding problem).

(b) For nitrogen, $M \cong 28$ amu, and with $\gamma = 1.4$ we obtain $dT/dz \cong -9.4^\circ\text{C}/\text{km}$, in reasonable agreement with experiment.

(c) From (68), $pV^\gamma = \text{constant}$, follows $p \propto V^{-\gamma}$. But $\rho = MN/V$, hence $p \propto \rho^{-\gamma}$.

To understand the instability of the atmosphere if the temperature gradient is steeper, consider two volumes of gas, V_1 and V_2 , at different heights $h_1 < h_2$ such that the local pressures are p_1 and p_2 . Choose the two volumes such that $p_1 V_1^\gamma = p_2 V_2^\gamma$. If the temperature gradient obeys (S1), both volumes contain the same mass. But if the higher-altitude volume is colder than required by (S1), it contains more mass than the lower volume. In this case the potential energy of the atmosphere is lowered by interchanging the two gas masses, expanding the lower volume isentropically until its volume equals that of the higher mass, and vice versa. The net volume work $-\int p dV$ required to interchange the two air masses is zero, hence the lowering of the potential energy is available to drive strong vertical convective flow. Tornadoes -- but not hurricanes -- work this way, although the details are complicated by the water content of the atmosphere.

On the other hand, if the temperature falls off less rapidly than according to (S2), any isentropic convection is strongly suppressed. This is the situation involved in the notorious inversion layer that can cause heavy air

pollution by preventing the dispersion of pollutants through vertical convection.

6.12) IDEAL GAS IN TWO DIMENSIONS

(a) Analogously to (3.59) and to (15),

$$\varepsilon_n = \frac{\hbar^2}{2M} \left(\frac{\pi}{L} \right)^2 (n_x^2 + n_y^2) = \frac{\hbar^2 \pi^2}{2MA} n^2, \quad n^2 = n_x^2 + n_y^2,$$

$$N = \sum f(\varepsilon_n) = \lambda \sum \exp(-\varepsilon_n/\tau) \quad .$$

The sum goes over $n_x > 0$ and $n_y > 0$ independently. We replace the sum by an integral

$$\sum \dots = (\pi/2) \int_0^\infty \dots n dn \quad .$$

If we substitute variables according to $x = \varepsilon_n/\tau = (\hbar^2 \pi^2 / 2MA\tau) n^2$, $dx = (\hbar^2 \pi^2 / MA\tau) n dn$, we obtain

$$N = \lambda \frac{\pi}{2} \frac{MA\tau}{\hbar^2 \pi^2} \int_0^\infty \exp(-x) dx = \lambda A n_Q^* \quad ,$$

where

$$n_Q^* = M\tau/2\pi\hbar^2 \quad (S1)$$

is the two-dimensional quantum concentration. Next:

$$\mu = \tau \log \lambda = \tau \log (n^*/n_Q^*) \quad , \quad n^* = N/A \quad , \quad (S2)$$

the same result as (18) in 3 dimensions, except for the replacement of n and n_Q by n^* and n_Q^* .

$$\begin{aligned} (b) \quad U &= \sum \varepsilon_n f(\varepsilon_n) = \lambda \sum \varepsilon_n \exp(-\varepsilon_n/\tau) \\ &= \lambda \tau^2 \frac{\partial}{\partial \tau} \sum \exp(-\varepsilon_n/\tau) \\ &= \lambda A \tau^2 (\partial n_Q^* / \partial \tau) = \lambda A n_Q^* \tau = N\tau \quad . \end{aligned}$$

(c) We noted that μ in (S2) had the same form as for a three-dimensional gas, except for the substitutions $n \rightarrow n^*$, $n_Q \rightarrow n_Q^*$. We may therefore obtain F as in (22) and (23):

$$F = N\tau[\log(n^*/n_Q^*) - 1] \quad .$$

From this, (33) and (S1)

$$\begin{aligned} \sigma &= -(\partial F / \partial \tau)_{A,N} = -N[\log(n^*/n_Q^*) - 1] - N\tau \\ &= N[\log(n_Q^*/n^*) + 2] \quad . \end{aligned}$$

6.13) GIBBS SUM FOR IDEAL GAS

$$\begin{aligned} \text{(a)} \quad \mathcal{Z} &= \sum_{ASN} \exp[(N\mu - \varepsilon_S)/\tau] = \sum_N \lambda^N \sum_S \exp(-\varepsilon_S/\tau) \\ &= \sum_N \lambda^N Z_N = \sum_N (\lambda n_Q V)^N / N! = \exp(\lambda n_Q V) \quad . \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad P(N) &= (\lambda n_Q V)^N / (N! \mathcal{Z}) = (\lambda n_Q V)^N \exp[-(\lambda n_Q V)] / N! \\ &= \langle N \rangle^N \exp(-\langle N \rangle) / N! \quad , \quad \text{with } \langle N \rangle = \lambda n_Q V \quad . \end{aligned}$$

$$\begin{aligned} \text{(c)} \quad \sum P(N) &= \exp(-\langle N \rangle) \sum \langle N \rangle^N / N! = \exp(-\langle N \rangle) \exp \langle N \rangle = 1 \quad . \\ NP(N) &= \langle N \rangle [\langle N \rangle^{N-1} / (N-1)!] \exp(-\langle N \rangle) = \langle N \rangle P(N-1) \quad ; \\ \sum_{N=0} NP(N) &= \sum_{N=1} NP(N) = \langle N \rangle \sum_{N=1} P(N-1) = \langle N \rangle \quad . \end{aligned}$$

6.14) IDEAL GAS CALCULATIONS

(a) The heat added during the isothermal expansion is

$$\begin{aligned} Q &= -W = \int_{V_1}^{2V_1} p dV = N\tau \int_{V_1}^{2V_1} dV/V = N\tau \log 2 \\ &= Nk_B \times 300 \text{ K} \times \log 2 = 1,729 \text{ J} \quad . \end{aligned}$$

There is no heat added during the isentropic expansion.

(b) From (66),

$$T_2(4V_1)^{\gamma-1} = T_1(2V_1)^{\gamma-1} ; T_2 = T_1 \times (1/2)^{\gamma-1} .$$

For a monatomic ideal gas $\gamma = 5/3$ and $\gamma-1 = 2/3$, hence

$$T_2 = 300 \text{ K} / 2^{2/3} = 189 \text{ K} .$$

(c) From (34), with $n_2 = n_1/2$

$$\begin{aligned} \Delta S &= k_B \Delta \sigma = N k_B [\log(n_Q/n_2) - \log(n_Q/n_1)] \\ &= N k_B \log 2 = 5.76 \text{ J K}^{-1} . \end{aligned}$$

6.15) DIESEL ENGINE COMPRESSION

From (66):

$$T_2 V_2^{\gamma-1} = T_1 V_1^{\gamma-1} ,$$

$$T_2 = T_1 \times (V_1/V_2)^{\gamma-1} = 300 \text{ K} \times 15^{0.4} = 886 \text{ K} = 613^\circ\text{C} .$$